MuonClip and Training Stability

Deep Dive into Kimi K3 — Chapter 13

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Foundation Books Project

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Goal: separate signal, intervention, and outcome; derive per-head QK-Clip for ordinary and latent attention; place it after Muon; make replicas agree; and read K2's evidence in scope.

Roadmap

Signals and the score path

Head-local QK-Clip

The feedback step and consensus

Evidence and diagnosis

Signals and the score path

A per-head signal on a bilinear score

- The scaled positive per-head maximum $s_{\max,t}^{(\ell,h)}$ is measured on W_{t-1}; global maxima and tail ratios are different statistics. Fixture: $(400, 80)$, tail $1/6$.

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- Muon's spectral breadth is a hypothesis chain to monitor, not a theorem. [REPORTED: K2 App. E; Moonlight App. D]

Head-local QK-Clip

One factor per score path

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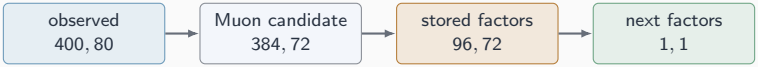
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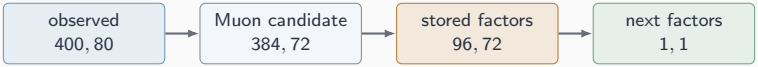
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- K2 MLA: $\sqrt{\gamma}$ on content query and key, γ on the rotary query, shared rotary key untouched. Replay 400 $\rightarrow$ 100; scaling the shared key would drag healthy head 2 from 80 to 50.

The feedback step and consensus

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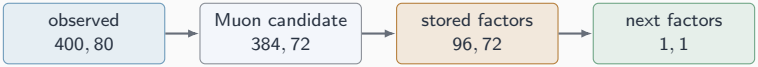


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- Replicas reduce an elementwise maximum per head before any commit: $(400, 80), (64, 120) \to (400, 120)$, factors $(1/4, 5/6)$ on both.

Evidence and diagnosis

Evidence, scope, and diagnosis

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Next: Chapter 14's Stable Sparse Pretraining.